

Integrated ERP System for Transportation Companies

Comprehensive Functional Scope for Developers

Prepared For	Development Team
Document Purpose	Define the full functional scope of a centralized ERP platform for transportation companies.
Target Users	Operations teams, fleet teams, finance, HR, management, technical support, and system integrators.
Core Scope	Operations, fleet, HR, payroll, accounting, tracking, camera monitoring, and internal display control.

This document outlines the required modules and key functions for a fully integrated ERP system designed specifically for transportation companies. The system should operate as one centralized platform connecting operational control, fleet readiness, human resources, accounting, monitoring technologies, and executive reporting.

1. System Overview

The platform is intended to manage the full operational cycle of transportation companies, from customer contracts and trip planning to driver assignment, bus tracking, maintenance, payroll, invoicing, and management reporting. It must provide live operational visibility, strong financial control, user permissions, and integration with smart devices such as GPS trackers, onboard cameras, and passenger display screens.

2. Core Modules Summary

No.	Module	Primary Objective
1	Operations Management	Plan, assign, execute, and monitor trips and daily operating orders.
2	Fleet Management	Manage buses, readiness, documents, and usage history.
3	Human Resources and Drivers	Maintain employee records, assignments, attendance, and performance.
4	Customers, Contracts, and Sales	Manage clients, quotations, contracts, and service orders.
5	Accounting and Finance	Control invoicing, expenses, receivables, payables, and profitability.
6	Maintenance and Workshop	Track preventive maintenance, breakdowns, work orders, and spare parts.
7	Fuel and Operating Expenses	Record fuel transactions and other route-related expenses.
8	Accidents, Violations, and Claims	Manage safety events, traffic violations, and insurance claims.
9	Payroll and Benefits	Process payroll, allowances, commissions, and deductions.
10	Reporting and Dashboards	Provide KPIs, operational reports, and executive insights.
11	External Integrations	Connect with GPS, WhatsApp, invoicing, and government platforms.
12	Access and Security	Control permissions, logs, approvals, backups, and data security.
13	Bus Tracking	Track buses in real time and analyze routes, delays, and movements.
14	Bus Camera Monitoring	Monitor live camera feeds and archive video linked to buses and trips.
15	Internal Bus Display Control	Manage onboard passenger screens and remote content updates.

3. Operations Management Department

- Create and manage single trips, scheduled trips, and trip cycles.
- Define service types such as staff transport, visitor transport, contract transport, Hajj and Umrah transport, and private transport.
- Enter trip data including date, time, origin, destination, route, client, and supervisor.
- Assign buses and drivers according to availability and operational readiness.
- Monitor trip execution in real time and manage trip status such as scheduled, active, completed, postponed, or canceled.
- Record bus condition before and after each trip and generate operational reports.

4. Fleet Management Department

- Register vehicle data including plate number, serial number, chassis number, model, type, and seating capacity.
- Monitor bus readiness and current status such as available, on trip, under maintenance, or inactive.
- Manage all official vehicle documents including registration, insurance, inspection certificates, and permits.
- Generate alerts before document expiration and maintain full operational and maintenance history.
- Link each bus to drivers, contracts, routes, and mileage records.

5. Human Resources and Drivers Management Department

- Maintain complete employee and driver profiles including ID, driving license, residence permit, contract details, and joining date.
- Manage attendance, shifts, leave, loans, penalties, and end-of-service records.
- Assign drivers according to routes, contracts, and operating plans.
- Calculate salaries, allowances, incentives, commissions, and deductions.
- Evaluate employees and drivers based on punctuality, discipline, violations, and customer feedback.

6. Customers, Contracts, and Sales Department

- Register individual clients, corporate clients, government entities, and external agents.
- Manage contracts, agreements, quotations, and approved pricing structures.
- Register bookings and service orders and link them to contracts and planned trips.
- Issue invoices and send schedules, quotations, and invoices through email or WhatsApp.
- Track customer history, revenue, and business activity.

7. Accounting and Finance Department

- Issue invoices linked to trips, services, and contracts.
- Record revenues, expenses, journal entries, receivables, payables, and financial transactions.
- Calculate trip cost, contract cost, and profitability.
- Manage cash, banks, vouchers, outstanding invoices, and collection follow-up.
- Generate financial statements, cost reports, and e-invoicing outputs compliant with local regulations.

8. Maintenance and Workshop Department

- Schedule preventive maintenance based on time or mileage.
- Record emergency breakdowns and create internal or external maintenance work orders.
- Register fault type, workshop, spare parts, cost, and downtime.
- Monitor vehicle status during workshop periods until return to operation.
- Manage spare parts inventory and generate maintenance cost and downtime reports.

9. Fuel and Operating Expenses Department

- Record fuel transactions including date, station, quantity, and cost.
- Link each fuel record to the bus, driver, and trip.
- Capture odometer readings and compare actual against expected fuel consumption.
- Manage fuel cards and analyze fuel costs by vehicle, route, driver, or contract.
- Track other operating expenses such as tolls, parking, and emergency route expenses.

10. Accidents, Violations, and Claims Department

- Record accident details including date, time, location, bus, driver, and related authorities.
- Attach official reports, photos, and supporting documents.
- Register traffic violations, assign responsibility, and follow up on payments.
- Manage insurance claims, settlement status, and compliance review.
- Generate safety and risk reports by bus, route, or driver.

11. Payroll and Employee Benefits Department

- Process monthly payroll for drivers, employees, and administrative staff.
- Calculate allowances, overtime, commissions, bonuses, and deductions.
- Link payroll calculations to attendance, violations, advances, or penalties.
- Issue salary statements and payroll summaries.
- Generate workforce cost reports by department, contract, or branch.

12. Reporting and Executive Dashboards Department

- Provide executive dashboards for top management.
- Generate operational, financial, HR, fleet, and driver performance reports.
- Provide KPI tracking for trip punctuality, utilization, safety, revenue, costs, and outstanding balances.
- Export reports in Excel and PDF formats.

13. External Integrations Department

- Integrate with GPS tracking systems for live vehicle location and route monitoring.
- Integrate with WhatsApp for trip orders, operational alerts, quotations, and invoices.
- Integrate with e-invoicing systems and available government platforms.
- Integrate with attendance systems and smart operational devices.

14. User Access, Permissions, and Security Department

- Create user roles and permissions based on department and job title.
- Separate access rights between operations, HR, finance, maintenance, and executive management.
- Maintain audit trails, system activity logs, backups, approval workflows, and data security controls.

15. Bus Tracking Department

- Track all buses in real time through GPS on interactive maps.
- Display current location, movement status, stop duration, departure time, arrival time, and delays.
- Compare the actual route against the planned route and generate alerts for route deviations or unauthorized stops.
- Monitor vehicle speed, distance traveled, operating hours, and route history.
- Integrate tracking data with operations, maintenance, accounting, and reports.

16. Bus Camera Monitoring Department

- Connect front, rear, and interior bus cameras to the central system.
- Monitor live camera feeds and archive video footage for investigations and service review.
- Generate smart alerts for unsafe behaviors such as phone use, driver drowsiness, harsh braking, or unusual activity.
- Link camera footage to buses, drivers, and trips and control permissions for access to live and recorded video.
- Monitor camera health, connectivity, and maintenance status.

17. Internal Bus Display Control Department

- Manage all internal passenger display screens from the central system.
- Display trip number, destination, driver name, estimated arrival time, service notices, and safety instructions.
- Update content remotely based on route, client, trip type, or operational plan.
- Integrate screen content with live trip and tracking data for automatic updates.
- Monitor screen performance, faults, and user permissions for content management.

18. Development Considerations

The system should be developed as a modular ERP platform with a centralized database and role-based access control. Each module must share master data consistently, especially buses, drivers, employees, clients, trips, contracts, and financial references. The preferred architecture should support future API integrations, mobile connectivity, real-time notifications, document attachments, audit logs, and scalable dashboard reporting.

19. Recommended Output for Phase 1

- Core master data setup for buses, drivers, employees, clients, routes, and contracts.
- Operational trip planning and assignment workflows.
- Finance, invoicing, receivables, and expense recording.

- HR, attendance, payroll, and commission calculation.
- Live bus tracking and event alerts.
- Camera monitoring and internal screen control integrations.
- Executive dashboards and printable reports.